# Assumption Mining: Extracting and Updating Implicit Requirements in LLM Code Generation

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## Abstract

Large language models (LLMs) have demonstrated strong capabilities in natural language-to-code generation, yet they frequently rely on unstated assumptions when user specifications are incomplete or ambiguous. These hidden assumptions—about input formats, boundary conditions, error handling, domain constraints, or intended semantics—remain implicit in generated code, making the resulting programs brittle, difficult to validate, and prone to misalignment with user intent.

This paper introduces *Assumption Mining*, a framework that enables LLMs to proactively surface, structure, and update implicit requirements during code generation. Instead of producing code alone, our approach augments generation with an explicit assumption layer that captures inferred constraints and design decisions in a formalized representation. We further propose an interactive interface that allows users to inspect, confirm, or modify extracted assumptions. When an assumption is flipped or revised, the system incrementally regenerates and updates the corresponding code while preserving unaffected components.

Our key contributions are: (1) a novel hidden-assumption generation framework integrated into LLM-based code synthesis; (2) an assumption–code dependency mechanism that supports traceable and controllable code updates; and (3) AmbigBench, a benchmark for evaluating hidden assumption extraction under ambiguous and underspecified prompts, including metrics for assumption quality, consistency, and code adaptability. We evaluate AssumptionMiner on AmbigBench across four dimensions: assumption extraction quality, assumption-to-code dependency accuracy, code adaptability after assumption revision, and developer experience in a controlled user study. Our results suggest that making implicit reasoning explicit is a critical step toward trustworthy, interactive LLM-based software engineering.

## CCS Concepts

• **Software and its engineering** → *Formal methods*; • **Computing methodologies** → **Natural language processing**.

## Keywords

LLM code generation, assumption mining, requirements engineering, implicit requirements, program synthesis, developer tools

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*ICSE '26, Somewhere, Someplace*
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ACM ISBN 978-X-XXXX-XXXX-X/XX/XX
https://doi.org/10.1145/XXXXXXX.XXXXXXX

**ACM Reference Format:**
Anonymous Author(s). 2026. Assumption Mining: Extracting and Updating Implicit Requirements in LLM Code Generation. In *Proceedings of 48th International Conference on Software Engineering (ICSE '26)*. ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/XXXXXXX.XXXXXXX

## 1 Introduction

Modern software development increasingly relies on large language models (LLMs) to accelerate code generation [? ? ]. Developers write natural-language prompts and receive executable code within seconds. Yet real-world prompts are rarely complete specifications: they leave open questions about input formats, boundary conditions, error-handling policies, domain constraints, and intended semantics. When an LLM fills these gaps, it silently commits to *hidden assumptions*—specific design choices that were never stated in the prompt [? ].

These hidden assumptions are a pervasive source of misalignment. The generated code may pass unit tests while violating project conventions, security policies, or the developer's actual intent—a class of defect we term *working-but-wrong*. Our empirical analysis of **[NUM: X]** real-world LLM-generated code artifacts found that **[NUM: X%]** contained at least one latent hidden assumption that could not be detected without manual inspection. The core difficulty is that existing tools give developers *no visibility* into what the LLM assumed. Prompt engineering guides [? ] advise more detailed prompts but cannot enumerate what was left out. Clarification-question systems [? ] ask questions before generation but do not link answers to specific code regions. Crucially, these approaches require developers to *know in advance what to ask*; AssumptionMiner instead *diagnoses* which decisions the LLM already made, surfacing them as revisable records grounded in the actual generated code. Self-repair methods [? ? ] iterate on test failures but cannot repair silent assumptions that produce passing-but-wrong code.

We present AssumptionMiner, a framework that makes hidden assumptions a *first-class output* of LLM code generation. Rather than extracting assumptions after the fact, AssumptionMiner augments the generation step itself: the backbone LLM jointly produces executable code *and* an explicit *assumption layer*—a structured, formalized list of every inferred constraint and design decision embedded in the generated code. Developers inspect, confirm, or modify individual assumptions through an interactive interface (§4); when an assumption is flipped or revised, AssumptionMiner incrementally regenerates only the code regions governed by that assumption while preserving all unaffected components.

**Contributions.**

(1) **Hidden-assumption generation framework** (§3): a novel approach that integrates assumption surfacing directly into LLM-based code synthesis, producing code and a formalized assumption layer in a single augmented generation step, making implicit LLM reasoning explicit and inspectable.

[FIGURE: Teaser figure: LEFT — developer submits ambiguous auth prompt; LLM generates code while silently assuming MD5 hashing, in-memory tokens, no rate-limiting. Misalignment discovered only at code review. RIGHT — AssumptionMiner augments the same generation step to jointly produce code + assumption layer (A1–A5). Developer confirms A1, flips A2 to bcrypt; incremental regeneration updates only the affected region.]

**Figure 1: Hidden assumptions in LLM code generation, before and after AssumptionMiner. Left: An ambiguous prompt yields code with five undisclosed assumptions (red) about password storage, session format, error policy, persistence, and rate-limiting—invisible until code review. Right: AssumptionMiner augments the same generation step to jointly produce code *and* an explicit assumption layer (A1–A5). The developer confirms three assumptions and flips A2 (MD5 → bcrypt); AssumptionMiner incrementally regenerates only the affected code region.**

(2) **Assumption–code dependency mechanism** (§3.4, §3.5): a two-pass dependency graph—first LLM-based candidate identification, then AST-level refinement—linking each assumption to the code regions it governs, enabling traceable and controllable incremental code updates when assumptions are flipped or revised.
(3) **AmbigBench benchmark** (§5): **[TODO: NUM]** programming tasks under ambiguous and underspecified prompts, annotated with expert-labeled hidden assumptions and evaluated with metrics spanning assumption quality, consistency, and code adaptability.

**Paper organization.** §2 defines implicit assumptions and motivates the problem with a running example. §3 describes the AssumptionMiner architecture. §4 presents the developer interface. §5 introduces AmbigBench. §6 reports experimental results. §7 surveys related work. §8 discusses threats to validity. §9 concludes.

# 2 Motivation & Background

## 2.1 Formal Definition

Let $p$ be a natural-language prompt (potentially ambiguous or underspecified) and $c = \mathcal{G}(p)$ be the code produced by an LLM $\mathcal{G}$. We define a *hidden assumption* as follows.

**Definition 1 (Hidden Assumption).** A *hidden assumption* $a$ of $(p, c)$ is a proposition that:

(i) is not stated in $p$ (arising from ambiguity or underspecification),
(ii) is instantiated in $c$ (i.e., $c$ behaves as if $a$ holds), and
(iii) has at least one alternative instantiation that satisfies $p$ but would produce a meaningfully different $c$.[a]

[a] We operationalize "meaningfully different" as producing distinct observable behavior on at least one input not excluded by $p$. For example, using MD5 vs. bcrypt for password hashing yields different outputs for any password string, satisfying this condition.

The set of hidden assumptions of $(p, c)$ is denoted $\mathcal{A}(p, c)$. In AssumptionMiner, $\mathcal{A}(p, c)$ is produced *during* generation as a formalized *assumption layer* co-output alongside $c$. Each assumption $a_i \in \mathcal{A}(p, c)$ is characterized by: (i) its *category* $\kappa(a_i)$ (see §2.3), (ii) a natural-language *description* $d(a_i)$, (iii) the *code regions* $R(a_i) \subseteq c$ it governs, and (iv) a set of *alternatives* $\mathrm{Alt}(a_i)$ representing choices the LLM could have made differently.

## 2.2 Running Example

We use the following prompt as a running example throughout the paper:

> *"Implement a user authentication system that checks credentials and returns a session token."*

A typical LLM-generated implementation embeds at least five implicit assumptions, which AssumptionMiner labels $A_1$ –$A_5$:

$A_1$ **Password storage:** Passwords are stored as MD5 hashes (weak; SHA-256 or bcrypt preferred).
$A_2$ **Session format:** Tokens are UUID v4 strings stored in-memory; no expiry logic.
$A_3$ **Error handling:** Invalid credentials return a bare boolean `False`; no structured error or HTTP status code.
$A_4$ **Persistence:** User records are held in a Python dict; no database or file persistence.
$A_5$ **Rate-limiting:** No brute-force protection is implemented.

**Table 1: Taxonomy of implicit assumptions in LLM-generated code. *Freq.* is percentage of benchmark tasks containing $\geq 1$ instance.**

| ID | Category | Description | Freq. |
|---|---|---|---|
| T1 | Security | Auth, crypto, input validation, access control | [NUM: X%] |
| T2 | Data Format | Types, serialization, encoding, nullability | [NUM: X%] |
| T3 | Error Policy | Exceptions, return values, HTTP codes | [NUM: X%] |
| T4 | Persistence | Storage backend, schema, consistency guarantees | [NUM: X%] |
| T5 | Performance | Caching, indexing, pagination, concurrency | [NUM: X%] |
| T6 | Architecture | Layer boundaries, API style, dependency choices | [NUM: X%] |

These five assumptions are silently embedded in 38 lines of the generated code (see fig. 1, right). Without a tool like AssumptionMiner, the developer must manually audit the code to discover them—a task our user study (§6.5) shows takes on average [NUM: X] minutes, vs. [NUM: X] minutes with AssumptionMiner.

*Second Example: An Algorithmic Function.* To demonstrate that hidden assumptions pervade even short algorithmic code, consider a prompt typical of competitive programming or coding interviews:

> *"Given a list of integers and a target value, find two numbers that add up to the target."*

A hash-map solution generated by GPT-4o in seven lines embeds four hidden assumptions invisible in the prompt:

$B_1$ **Return type (T2):** The function returns a pair of *indices* (e.g., `[1, 3]`), not the *values* themselves—a choice the prompt does not specify.

$B_2$ **Uniqueness (T3):** Exactly one valid pair is assumed; the first match is returned without checking for additional solutions.

$B_3$ **No-solution policy (T3):** An empty list `[]` is returned when no pair sums to the target—not `None`, not a raised `ValueError`.

$B_4$ **Space–time trade-off (T5):** An $O(n)$-space hash map is chosen over an $O(n^2)$ brute-force scan or an $O(n \log n)$ sort-and-scan approach.

Unlike the authentication example, these assumptions arise in seven lines of purely algorithmic code with no external dependencies, confirming that hidden assumptions are not confined to system-design tasks: they emerge whenever a prompt leaves behavioral or design questions open, regardless of code complexity.

## 2.3 Assumption Taxonomy

Based on open coding of [NUM: X] LLM-generated code samples following an iterative card-sorting protocol [? ], we derive a taxonomy of 6 assumption categories (table 1).

[FIGURE: Architecture diagram: (1) Prompt Ingestion → (2) Code Generator (LLM) → (3) Assumption Extractor (LLM + structured prompt) → AssumptionRecord store → (4) Dependency Mapper → code annotation → (5) Incremental Regenerator. Arrows show data flow; dashed box for developer interaction loop.]

**Figure 2: AssumptionMiner system architecture. Boxes denote components; arrows denote data flow. The dashed region marks the developer interaction loop.**

# 3 The AssumptionMiner Framework

## 3.1 Architecture Overview

AssumptionMiner consists of four loosely-coupled components connected by a structured data model (fig. 2).

**C1 — Prompt Ingestion.** Receives the developer's natural-language prompt $p$, optionally enriched with project-level context (language, framework, coding style guide).

**C2 — Code Generator.** Issues $p$ to the backbone LLM and produces the initial code artifact $c$. Any instruction-tuned LLM (GPT-4o, Claude 3.5, Gemini 1.5) may serve as the backbone.

**C3 — Assumption Extractor.** Given $(p, c)$, applies a structured extraction prompt (§3.2) to enumerate $\mathcal{A}(p, c)$ as a list of *AssumptionRecord* objects (§3.3).

**C4 — Dependency Mapper.** Links each assumption to the code regions it governs via a lightweight AST-level analysis (§3.4).

**C5 — Incremental Regenerator.** When the developer revises an assumption $a_i$, regenerates only the code regions $R(a_i)$ while preserving all other regions (§3.5).

## 3.2 Assumption Extraction

The core of AssumptionMiner is a two-phase extraction prompt (listing 1). Phase 1 asks the LLM to reason about what choices were *made* in $c$ that go beyond what $p$ requires—a chain-of-thought [? ] reasoning step. Phase 2 asks the model to format its discoveries into

```
SYSTEM: You are an expert software engineer
    auditing LLM-generated
code for implicit design decisions.

USER:
### Original Prompt
{PROMPT}

### Generated Code
{CODE}

### Task
1. REASON: List every design decision in the code
    that is NOT
   required by the prompt. For each, name a
       realistic alternative.
2. FORMAT: Return a JSON array of AssumptionRecord
     objects.
   Schema: {schema}
   Return ONLY valid JSON after the tag [RECORDS].

[RECORDS]
```

**Listing 1: Assumption extraction prompt template (abbreviated). `{PROMPT}` and `{CODE}` are replaced at runtime.**

```
{
  "id":          "string",        // e.g. "A1"
  "category":    "T1|T2|T3|T4|T5|T6",
  "description": "string",       // natural-
      language statement
  "rationale":   "string",       // why LLM made
      this choice
  "alternatives": ["string"],    // >= 1
      alternatives
  "code_refs": [{                // dependency info
       (C4 populates)
    "file":       "string",
    "start_line": "int",
    "end_line":   "int",
    "ast_node":   "string"       // optional AST
        node type
  }],
  "confidence": "float",         // [0, 1]
      extraction confidence
  "severity":   "low|medium|high"  // LLM-assigned
      ; human may override
}
```

**Listing 2: AssumptionRecord JSON schema.**

the structured *AssumptionRecord* schema, enabling downstream programmatic processing.

## 3.3 AssumptionRecord Schema

Each extracted assumption is stored as an *AssumptionRecord* (listing 2). The schema is designed for both human readability and programmatic processing by C4 (dependency mapping) and C5 (incremental regeneration).

**[FIGURE: Dependency graph for running example: nodes A1–A5 on left, AST nodes (FunctionDef: hash_password, Assign: token=uuid4(), Return: False, Dict: users, ...) on right. Directed edges with confidence scores.]**

**Figure 3: Assumption-code dependency graph for the running example. Left: assumption nodes $A_1$ –$A_5$. Right: AST nodes in the generated code. Edge weight = dependency confidence.**

## 3.4 Assumption-Code Dependency Mapping

To support targeted incremental regeneration, AssumptionMiner must link each assumption to the code regions it governs. This is non-trivial: a single assumption (e.g., *password hashing algorithm*) may manifest across multiple non-contiguous lines and functions.

We construct an *Assumption-Code Dependency Graph* $G = (V_a \cup V_c, E)$ (fig. 3), where $V_a$ is the set of assumptions, $V_c$ is the set of AST nodes in $c$, and $(a_i, n_j) \in E$ if assumption $a_i$ governs node $n_j$. Edge weights reflect confidence.

Edge construction proceeds in two passes:

(1) **LLM pass.** The extraction prompt (listing 1) requests natural-language descriptions of relevant code regions. We parse these with a token-overlap heuristic: we tokenize both the LLM's region description and each code identifier, then retain AST nodes whose identifier tokens share a Jaccard similarity $\geq 0.25$ with the description tokens (threshold chosen by pilot on 20 held-out tasks).

(2) **AST refinement pass.** We parse $c$ with the `tree-sitter` library and perform a bottom-up traversal from each candidate leaf node, stopping at the deepest ancestor that encloses *all* candidate nodes for that assumption; this yields the minimal enclosing sub-tree.

## 3.5 Incremental Regeneration

When a developer revises assumption $a_i$ to a new value $a_i'$, AssumptionMiner identifies $R(a_i) = \{n \in V_c : (a_i, n) \in E\}$ and constructs a targeted regeneration prompt containing only the affected sub-tree, the revised assumption, and the surrounding context (at most $k$ lines above/below each region, $k = 5$ in our experiments). The

remainder of $c$ is left unchanged. This approach is analogous to *in-place surgical editing* [? ], but guided by explicit assumption metadata rather than test-failure signals.

## 4 Interactive Interface

The AssumptionMiner interface (fig. 4) is a web-based IDE panel implemented in React that augments a standard code editor. Developers interact through three primary actions:

**Inspect.** Each AssumptionRecord is rendered as a card in the right panel, color-coded by severity (red = high, amber = medium, green = low). Hovering a card highlights the corresponding code regions in the editor; clicking an assumption scrolls to its primary location.

**Edit.** Developers may modify the *description* field or select an alternative from the *alternatives* dropdown. Custom alternatives may be entered as free text. Changes are staged but not yet applied.

**Regenerate.** Pressing *Regenerate* triggers C5 for all staged edits. AssumptionMiner performs topological ordering over the dependency graph to process interdependent assumptions in a safe sequence, then presents a diff view for developer approval before committing changes.

## 5 AmbigBench Benchmark

To enable systematic, reproducible evaluation of assumption mining methods, we construct AmbigBench (table 2).

**Task collection.** AmbigBench comprises **[TODO: NUM]** programming tasks drawn from three sources: **[NUM: X]** tasks from HumanEval [? ], **[NUM: X]** tasks from MBPP [? ], and **[NUM: X]** novel tasks designed to stress-test specific assumption categories (particularly T1–Security and T4–Persistence, which are underrepresented in existing benchmarks).

**Annotation.** Each task was independently annotated by **[NUM: X]** graduate students with > 2 years of software development experience. Annotators labeled all assumptions they could identify per Definition 1, categorized them by the taxonomy in §2.3, and specified the corresponding code regions. Inter-annotator agreement (Cohen's $\kappa$) was **[NUM: $\kappa$]** after one calibration round. Disagreements were resolved by a senior author.

**Benchmark statistics** are summarized in table 2.

**Metrics.** We define five evaluation metrics over $\mathcal{A}(p, c)$ and a reference set $\mathcal{A}^*(p, c)$:

*AR* **(Assumption Recall):** $|\mathcal{A} \cap \mathcal{A}^*| \,/\, |\mathcal{A}^*|$

*AP* **(Assumption Precision):** $|\mathcal{A} \cap \mathcal{A}^*| \,/\, |\mathcal{A}|$

$AF_1$ **(Assumption $F_1$):** Harmonic mean of *AR* and *AP*

*AS* **(Assumption Specificity):** Mean BLEU-4 score of generated descriptions against reference descriptions for matched assumptions. We adopt BLEU-4 as a lexical-overlap proxy: assumption descriptions are short, factual sentences ("passwords stored as MD5 hashes") where n-gram precision captures whether the key constraint tokens appear, unlike open-ended fluency tasks where BLEU is less appropriate. We acknowledge this does not capture paraphrastic equivalence; semantic similarity scores (e.g., BERTScore) are a planned extension.

*CAS* **(Category-Adjusted Specificity):** *AS* weighted by inverse category frequency (penalizes easy categories)

**Table 2: AmbigBench benchmark statistics (pilot $n = 10$; target $n = 150$). Full benchmark in preparation; pilot figures shown in parentheses.**

| Metric | Value |
|---|---|
| Total tasks (target) | 150 (pilot: 10) |
| Tasks from HumanEval | 50 (pilot: 0) |
| Tasks from MBPP | 50 (pilot: 0) |
| Novel tasks | 50 (pilot: 10) |
| Total ground-truth assumptions | **[NUM: X]** (pilot: 39) |
| Mean assumptions per task | **[NUM: X]** (pilot: 3.9) |
| Median assumptions per task | **[NUM: X]** (pilot: 4.0) |
| T1 Input Validation | **[NUM: X]** (pilot: 5, 12.8%) |
| T2 Data Format | **[NUM: X]** (pilot: 13, 33.3%) |
| T3 Error Policy | **[NUM: X]** (pilot: 8, 20.5%) |
| T4 Persistence | **[NUM: X]** (pilot: 4, 10.3%) |
| T5 Performance | **[NUM: X]** (pilot: 7, 17.9%) |
| T6 Security | **[NUM: X]** (pilot: 2, 5.1%) |
| Mean lines of code per task | **[NUM: X]** (pilot: 10.0) |
| Inter-annotator agreement ($\kappa$) | **[NUM: X]** (pilot: N/A) |

## 6 Evaluation

We evaluate AssumptionMiner with four research questions:

**RQ1 — Extraction Quality**

How accurately does AssumptionMiner identify and characterize implicit assumptions compared to expert annotations?

**RQ2 — Dependency Accuracy**

How precisely does the dependency mechanism link assumptions to code regions?

**RQ3 — Code Adaptability**

When assumptions are revised, how accurately does incremental regeneration produce correct code updates?

**RQ4 — User Study**

Does AssumptionMiner reduce user effort and improve satisfaction compared to a standard LLM interface?

### 6.1 Experimental Setup

**Backbone LLM.** We use GPT-4o (`gpt-4o-2024-05-13`) for all AssumptionMiner components unless otherwise stated. Ablations with Claude 3.5 Sonnet (`claude-3-5-sonnet-20241022`) and Gemini 1.5 Pro (`gemini-1.5-pro-002`) are reported in table 3.

**Baselines.**

- *Direct Generation (DG):* Vanilla LLM generation; no assumption mining. Assumptions derived by having annotators label the output post-hoc.
- *Comment Extraction (CE):* Extracts design decisions from code comments and docstrings only.

[FIGURE: UI mockup (annotated): LEFT PANEL — prompt input + generated code with highlighted assumption regions (color-coded by severity). RIGHT PANEL — AssumptionRecord list; each card shows id, category, description, alternatives dropdown, severity badge. BOTTOM BAR — "Accept All", "Edit", "Regenerate" buttons. Callout boxes A1–A5 pointing to relevant code lines.]

**Figure 4: AssumptionMiner interactive interface. (1) Highlighted code regions correspond to extracted assumptions. (2) The assumption panel lets developers inspect and edit each assumption record. (3) The revision panel applies changes through incremental regeneration.**

- *Clarification Questions (CQ):* Presents the developer with open-ended clarification questions before generation, then mines the conversation [? ].
- *Chain-of-Thought Extraction (CoT):* Prompts the LLM to reason step-by-step about design decisions without our structured schema [? ].

**Dependency baselines (RQ2):**

- *Keyword Heuristic (KH):* Token-overlap between assumption description and code tokens; no AST information.
- *Full-File Dependency (FF):* Trivially assigns every assumption to the entire file (upper bound on recall, lower bound on precision).

**Regeneration baselines (RQ3):**

- *Full Regeneration (FR):* Re-generates the entire function from scratch with the revised assumption in the prompt.
- *In-Place LLM Edit (IPE):* Passes the full code + revision instruction to the LLM and asks it to apply the change [? ].
- *Manual Reprompt (MR):* Developer manually rewrites the prompt; LLM generates fresh code.

## 6.2 RQ1: Extraction Quality

table 3 reports *AR*, *AP*, $AF_1$, *AS*, and *CAS* for all methods on the full AmbigBench test set.

table 4 provides a per-category breakdown for the best configuration (AssumptionMiner with GPT-4o).

**Table 3: RQ1: Assumption extraction quality. Best results in bold; second-best underlined. † statistically significant over all baselines ($p < 0.05$, Wilcoxon signed-rank).**

| Method | *AR* | *AP* | $AF_1$ | *AS* | *C* |
|---|---|---|---|---|---|
| Direct Generation | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM |
| Comment Extraction | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM |
| Clarif. Questions | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM |
| CoT Extraction | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM |
| AssumptionMiner (GPT-4o) | [NUM: ] | [NUM: ] | [NUM: ]† | [NUM: ] | [NUM |
| AssumptionMiner (Claude 3.5) | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM |
| AssumptionMiner (Gemini 1.5) | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM |

**Table 4: RQ1 per-category breakdown for AssumptionMiner (GPT-4o).**

| Category | *AR* | *AP* | $AF_1$ | *AS* | *CAS* |
|---|---|---|---|---|---|
| T1 Security | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |
| T2 Data Format | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |
| T3 Error Policy | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |
| T4 Persistence | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |
| T5 Performance | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |
| T6 Architecture | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |
| **Overall** | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |

**Findings.** **[TODO: Summarize RQ1 findings: which categories are hardest, effect of structured schema vs. CoT, precision-recall trade-offs.]**

**Table 5: RQ2: Dependency accuracy on AmbigBench (pilot, $n = 10$ tasks, 39 assumptions). Accuracy = fraction with IoU $\geq$ 0.5. IoU = mean Intersection over Union. — = metric not applicable.**

| Method | Line-level | | Category breakdown (Acc.) | |
|---|---|---|---|---|
| | Acc. | IoU | Best cat. | Worst cat. |
| Keyword Heuristic | 0.513 | 0.547 | T4 (—) | T1 (—) |
| Full-File Dependency | 0.154 | 0.240 | — | — |
| AssumptionMiner (heuristic) | 0.333 | 0.346 | T4 (0.75) | T1 (0.00) |
| AssumptionMiner (LLM+AST) | [NUM: ] | [NUM: ] | [NUM: ] | [NUM: ] |

**RQ1 Summary**

**[TODO: 2–3 sentence answer to RQ1.]**

## 6.3 RQ2: Dependency Accuracy

We evaluate dependency link quality at two granularities: *line-level* and *AST-node-level*. Ground-truth links were annotated alongside AmbigBench assumptions. Results are reported in table 5.

**Findings.** On our pilot benchmark ($n = 10$ tasks, 39 expert-annotated assumptions), the tree-sitter heuristic achieves 33.3% accuracy (mean IoU = 0.35) at the default 0.5 threshold. The keyword-heuristic baseline (KH) achieves a higher 51.3% on this pilot, primarily because many ground-truth regions are single-line; the tree-sitter mapper returns larger AST sub-trees that dilute IoU. The LLM-guided pass (full AssumptionMiner, §3.4) is expected to close this gap by using semantic descriptions rather than surface keywords; results on the complete 150-task AmbigBench are pending. T4 (Persistence) is the easiest category to localise (0.75 accuracy)—variable declarations and import statements provide reliable surface cues. T1 (Input Validation) is hardest (0.00)—input handling is spread across multiple lines and rarely associated with distinctive identifiers.

**RQ2 Summary (pilot)**

The heuristic dependency mapper localises 33% of assumptions to the correct code region (IoU $\geq$ 0.5); the LLM-guided pass is hypothesised to improve this by leveraging semantic descriptions. T4 (Persistence) assumptions are easiest to localise; T1 (Input Validation) are hardest due to dispersed implementation.

## 6.4 RQ3: Code Adaptability

We simulate assumption revision by modifying one assumption per task (selecting the highest-severity assumption) and measuring whether the regenerated code correctly reflects the change. Quantitative results appear in table 6.

**Metrics.**

- *Pass Rate*: percentage of tasks where revised code passes all provided unit tests.
- *Edit Distance*: normalized Levenshtein edit distance between original and revised code (lower = more targeted).
- *Latency*: wall-clock time for the regeneration step in seconds.

**Table 6: RQ3: Code adaptability when one assumption is revised.**

| Method | Pass Rate | Edit Dist. | Latency (s) |
|---|---|---|---|
| Manual Reprompt | [NUM: ] | [NUM: ] | [NUM: ] |
| Full Regeneration | [NUM: ] | [NUM: ] | [NUM: ] |
| In-Place LLM Edit | [NUM: ] | [NUM: ] | [NUM: ] |
| AssumptionMiner (Incremental) | [NUM: ] | [NUM: ] | [NUM: ] |

**Findings. [TODO: Summarize RQ3 findings: does incremental beat full regen on pass rate? latency advantage? are there failure modes?]**

**RQ3 Summary**

**[TODO: 2–3 sentence answer to RQ3.]**

## 6.5 RQ4: User Study

**Design.** We conducted a controlled within-subjects study with **[TODO: NUM]** developers (**[NUM: X]** industry, **[NUM: X]** graduate students; **[NUM: X]**+ years of Python experience on average). Each participant completed **[NUM: X]** tasks under two conditions in counterbalanced order: *Baseline* (standard ChatGPT interface) and *AssumptionMiner*. Tasks were drawn from AmbigBench and required identifying and correcting at least one assumption.

**Measures.** *Task completion time* (minutes), *assumption coverage* (fraction of ground-truth assumptions correctly identified and acted upon, where "acted upon" means the participant explicitly accepted, rejected, or edited the corresponding assumption card before submitting their solution), and *SUS score* [? ] (system usability scale, 0–100).

fig. 5 presents the task-completion time and SUS distributions. **Findings. [TODO: Report median task time, median SUS, statistical significance. Quote representative qualitative comments from exit interview.]**

**RQ4 Summary**

**[TODO: 2–3 sentence answer to RQ4.]**

## 6.6 Error Analysis

fig. 6 illustrates six representative failure modes observed during evaluation. **[TODO: Describe each failure mode briefly and its frequency.]**

# 7 Related Work

## 7.1 LLM-Based Code Generation

Large language models have achieved strong performance on programming benchmarks such as HumanEval [? ], MBPP [? ], and SWE-bench [? ]. Self-repair approaches [? ? ] improve correctness by iterating on test failures, while self-planning methods [? ? ] decompose tasks before generation. *None of these works surface implicit assumptions or connect them to code regions for targeted revision.*

[FIGURE: Side-by-side boxplots: LEFT — task completion time (minutes) for Baseline vs. AssumptionMiner; RIGHT — SUS scores. Significance annotation bars (Wilcoxon, $p < 0.05$).]

**Figure 5: RQ4 user study results. Task-completion time in minutes (left) and SUS usability scores (right) for Baseline vs. AssumptionMiner. Boxes show IQR; whiskers show 1.5×IQR; $*p < 0.05$, $**p < 0.01$.**

## 7.2 Requirements Engineering & Ambiguity

A rich body of RE research addresses natural-language ambiguity detection [? ? ? ]. Classical approaches identify lexical or syntactic ambiguity but do not model the relationship between underspecified requirements and generated implementation decisions. *AssumptionMiner extends this line of work to the code-generation setting, where the artifact under analysis is executable code rather than a specification document.*

## 7.3 Traceability & Dependency Analysis

Traceability research [? ? ] links requirements artifacts to code using information-retrieval or ML techniques. Static analysis tools [? ? ] provide program-level dependency graphs. *AssumptionMiner combines semantic (LLM-based) dependency inference with lightweight AST analysis to produce the first assumption-to-code dependency graphs for LLM-generated artifacts.*

## 7.4 Interactive & Conversational Code Generation

Conversational repair tools [? ] use test-driven iteration, while tools like Copilot rely on implicit context from the surrounding file. User studies [? ? ] show that developers frequently struggle to identify *why* generated code is wrong. *AssumptionMiner directly addresses this gap by making implicit design decisions explicit and editable before they cause downstream failures.*

[FIGURE: 2×3 grid of qualitative error examples: false negatives (missed assumptions), false positives (hallucinated assumptions), wrong category, imprecise dependency, failed regeneration, cascading edit. Each cell shows abbreviated code snippet + assumption card.]

**Figure 6: Representative error cases across six failure modes.**

## 7.5 Specification Inference & Contract Mining

Automated specification mining [? ] and contract inference techniques extract program invariants and behavioral contracts from executions or static analysis. LLM-augmented testing tools such as CodaMosa [? ] and Fuzz4All [? ] leverage language models to guide coverage-driven exploration and oracle generation. Unlike AssumptionMiner, all of these techniques operate *post-hoc* on existing code: they cannot attribute discovered properties to specific prompt ambiguities, nor do they surface assumptions *during* generation for developer review. AssumptionMiner is complementary—its structured assumption layer could *seed* downstream invariant checkers with candidate properties, but its primary goal is interactive disambiguation at generation time rather than post-hoc property discovery.

## 7.6 Architectural Decision Records

Architectural Decision Records (ADRs) [? ? ] are lightweight documents that capture the rationale behind a design decision, its alternatives, and its consequences. The AssumptionRecord schema in AssumptionMiner is directly inspired by this tradition: both treat design decisions as first-class, addressable artifacts with explicit alternatives. The key distinction is automation: ADRs are authored manually by architects after the fact, whereas AssumptionMiner *generates* assumption records automatically during code synthesis, making the practice tractable for the routine, fine-grained decisions embedded in everyday LLM code generation.

### 7.7 Automated Program Repair

Automated Program Repair (APR) techniques [? ? ? ] generate patches that correct program failures, typically guided by failing tests or formal specifications. The incremental regeneration component of AssumptionMiner (C5) shares structural similarities with patch generation: both isolate a minimal code region and apply a targeted transformation. However, APR is *fault-driven*—triggered by observable failures—while AssumptionMiner's regeneration is *assumption-driven*, triggered by developer preference changes on code that may already pass all tests (the working-but-wrong regime that APR cannot address). This makes C5 closer in spirit to requirement-driven change propagation than to defect localization and repair.

## 8 Threats to Validity

We assess threats following Wohlin et al. [? ].

**Internal validity.** Annotation bias may affect AmbigBench ground truth: annotators were graduate students aware of the study goals. We mitigated this with calibration rounds and senior-author adjudication. LLM non-determinism (temperature > 0) means reported numbers reflect averaged runs; we report means over **[NUM: X]** runs per task.

*LLM backbone bias.* The assumption taxonomy and AmbigBench annotations were developed while using GPT-4o, and AssumptionMiner's extractor is also GPT-4o-based; evaluation on GPT-4o outputs thus carries a risk of circular self-consistency rather than true generalization. We partially mitigate this by (i) using held-out tasks not seen during taxonomy development, and (ii) reporting ablations with Claude 3.5 Sonnet and Gemini 1.5 Pro (see table 3), but full independence would require a backbone-agnostic benchmark constructed without any LLM assistance.

*Benchmark contamination.* HumanEval [? ] and MBPP [? ] tasks are part of GPT-4o's training data; the model may therefore generate predictable, stereotyped solutions whose assumptions are easier to extract, potentially inflating extraction accuracy relative to novel tasks. The **[NUM: X]** novel tasks in AmbigBench are designed to be out-of-distribution for this reason, and we report per-source breakdowns to allow readers to assess this effect.

**External validity.** AmbigBench tasks are drawn from Python; results may not generalize to statically typed languages (e.g., Rust, Java) where some assumptions are enforced by the type system. Industrial-scale codebases with multi-file context may expose additional assumption categories not in our taxonomy.

**Construct validity.** Our five metrics ($AR$, $AP$, $AF_1$, $AS$, $CAS$) operationalize assumption quality as human-judged matches; they cannot capture all nuances of semantic equivalence. The SUS instrument [? ] measures perceived usability, which correlates with but does not equal actual productivity gains.

**Conclusion validity.** The user study sample (**[TODO: NUM]** participants) is sufficient for detecting medium-to-large effect sizes ($d \geq 0.5$, power $\geq 0.8$) but may be under-powered for small effects. Although task order was counterbalanced across participants, within-subjects designs carry learning-effect risks: participants who encounter AssumptionMiner second may perform better on the Baseline condition due to improved familiarity with the task type itself, not the tool. We analyze order as a covariate in our statistical models to detect and control for this confound.

## 9 Conclusion

LLM code generation silently embeds a large number of implicit assumptions that developers have no systematic way to inspect or revise. These hidden decisions are a root cause of working-but-wrong code—code that passes tests yet violates developer intent.

We presented AssumptionMiner, a framework that mines implicit assumptions from LLM-generated code, surfaces them as structured *AssumptionRecord* objects linked to their governing code regions, and propagates developer-specified revisions through targeted incremental regeneration. Evaluated on our new AmbigBench benchmark, AssumptionMiner achieves **[NUM: X]** assumption $F_1$—a gain of **[NUM: +X pp]** over the strongest baseline—and **[NUM: X%]** pass rate after assumption revision, with **[NUM: X]**% lower latency than full regeneration. A user study confirms that developers complete assumption-inspection tasks **[NUM: X%]** faster and rate AssumptionMiner significantly higher on usability.

Future work includes extending AssumptionMiner to multi-file repositories, incorporating project-level style guides as assumption priors, and studying how assumption mining interacts with test generation tools. We release AmbigBench and the AssumptionMiner implementation at **[TODO: anonymized URL]**.